

MUHAMMAD MUSTAFA KHAN

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RESEARCH INTERESTS

Robotics and feedback control of physical systems that sense, process information, decide and act in real time. My research combines system identification, signal processing, control design, sensing, actuation and real-time embedded implementation, with a growing interest in learning-enabled control, Physical AI and embodied intelligence.

EDUCATION

Gwangju Institute of Science and Technology (GIST)

Sep 2025 – Aug 2027 (exp.)

M.S., Mechanical and Robotics Engineering · GPA 3.85 / 4.50

National University of Sciences and Technology (NUST)

Sep 2019 – Jun 2023

B.E., Mechanical Engineering · GPA 3.43 / 4.00

Selected graduate coursework: Robot Control · AI Experience Lab · Introduction to Self-Driving Cars · Electronic Circuit Analysis and Signal Processing · Intelligent Motor Integration Project · Control Engineering · Introduction to Mechatronics Design Fundamentals

RESEARCH EXPERIENCE

Graduate Researcher

Dynamics & Control Lab, School of Mechanical and Robotics Engineering, GIST

Sep 2025 – present

Sensing, identification and real-time embedded control of a high-speed AFM system

- Measured the open-loop frequency response of a custom-built contact-mode AFM system by swept-sine excitation about the contact operating point, and identified a fourth-order plant model with a 4.82 kHz dominant resonance.
- Designed the Z-axis PI controller by loop shaping on that measured response and implemented it on an STM32F407 at 18.4 kHz, driven from the ADC conversion-complete interrupt with anti-windup and piezo-drive saturation.
- Redesigned the AFM PCB, adding a buffered level shifter that eliminated a reference-voltage shift under applied input and a modulation-frequency amplifier stage, then integrated the piezo and nano-stage drive electronics and carried the revision through external fabrication.
- Quantified imaging fidelity reference-free on a 100 nm step-height standard by trace–retrace Pearson correlation and NRMSE, a measure of bidirectional consistency rather than ground-truth accuracy: $r \geq 0.977$ and $\text{NRMSE} \leq 7.7\%$ up to 5 Hz, falling to $r = 0.856$ and $\text{NRMSE} = 19.5\%$ at 20 Hz.
- Identified closed-loop tracking bandwidth as the principal limitation at higher scan rates and implemented notch compensation at the measured resonance on a higher-rate STM32 build; model-predictive control is being developed and evaluated against the identified model in simulation.

Research Assistant

Human System Lab, School of Mechanical and Manufacturing Engineering, NUST

Jun 2022 – Aug 2023

Real-time sEMG-driven interface for post-stroke upper-limb rehabilitation

- Developed a real-time sEMG acquisition and processing pipeline using all eight Myo armband channels at the device's fixed 200 Hz sampling rate, with 150 ms sliding windows, 73 % overlap, four time-domain features per channel and Z-score normalisation.
- Compared SVM, decision tree, XGBoost, CatBoost and gradient-boosting classifiers through cross-validation, then deployed the selected gradient-boosting model for continuous online gesture inference.
- Designed and implemented classifier-driven rehabilitation game interfaces with one, two and three degrees of freedom, including a custom three-DOF maze.
- Developed the experimental protocol and evaluated interface performance using Fitts' law metrics—index of difficulty and throughput—together with completion rate, overshoot, completion time, path efficiency and goniometric joint-angle measurements.
- Regressed completion time against index of difficulty and observed strong participant-level model fits, with $R^2 \geq 0.98$ across ten able-bodied participants and $R^2 \geq 0.88$ for two participants with chronic stroke; prepared the recorded signals for release as the MUSED-I dataset.

RESEARCH OUTPUTS

- **[Poster]** M. M. Khan, T. Abbas and K. H. Park, “Model-Based Digital Proportional-Integral Control for High-Speed Contact-Mode Atomic Force Microscopy Imaging,” *23rd IFAC World Congress*, Busan, Korea, 23–28 August 2026. Presented by M. M. Khan.
- **[Dataset]** M. M. Khan, M. Farhan, A. Shahzad, H. S. Qarni, A. Waris and O. Gilani, “MUSED-I: Multi-Gesture Surface sEMG Dataset for Stroke Rehabilitation,” *IEEE DataPort*, August 2023, doi:10.21227/04zq-yz45.

SELECTED ROBOTICS, CONTROL & MECHATRONICS PROJECTS

Voice-Coil Actuator Positioning System: Identification and Analog Feedback Control

GIST · Winter 2025–26

- Designed the actuator bench in SolidWorks, fabricated and assembled the hardware, and instrumented it with a PA51 power operational amplifier as a transconductance stage and laser-triangulation position feedback.
- Characterised the plant by open-loop frequency-response analysis and derived a mass–damper model with no restoring spring term.
- Designed, built and experimentally evaluated analog P, D, PD, PI, lead and lag compensators as op-amp circuits, verifying their measured responses against the predicted Bode behaviour; PI introduced excessive phase lag, so lead compensation was retained and the resulting closed-loop response validated experimentally.

REHABOTICS: Pneumatic Soft Robotic Glove for Hand Rehabilitation

NUST · 2022–2023

- Designed fibre-reinforced pneumatic bending actuators across three geometry iterations against requirements of sub-0.5 kg mass and 10–15 N flexion force, analysing stress distribution and deformation in ANSYS.
- Fabricated the actuators by soft lithography — two-part SolidWorks moulds 3D-printed in PLA, Ecoflex 00-30 cast under vacuum degassing, a buckram strain-limiting layer and fibre-thread reinforcement — achieving 120–180° of flexion per finger segment at 100 kPa.
- Integrated the pneumatic drive and master–slave control on a Raspberry Pi 4 with diaphragm pump, solenoid valves, regulator and pressure feedback, so that a gesture from the functional hand actuates the paretic hand, and validated range of motion experimentally.

Induction-Motor Condition Monitoring: Multisensor Acquisition and Autoencoder-Based Anomaly Detection

GIST · 2025

- Built an induction-motor monitoring rig instrumented with synchronised MPU-6050 vibration, DS18B20 temperature, ACS712 current and A3144 Hall-effect shaft-speed sensing on an Arduino, stabilising the current channel with a moving-average filter.
- Acquired a healthy-condition baseline dataset and implemented the Python signal-processing and windowed feature-extraction pipeline over the synchronised sensor streams.
- Evaluated candidate machine-learning approaches on that baseline and selected an autoencoder for unsupervised anomaly detection, deploying the trained model behind a real-time monitoring dashboard.

HONOURS & AWARDS

- **Graduate Research Excellence Award – Research Assistant Scholarship**, competitive departmental scholarship, Department of Mechanical and Robotics Engineering, GIST — December 2025.
- **NGIRI Research Funding**, Ignite / Government of Pakistan — competitive national undergraduate research grant awarded to the Final Year Project, July 2023.
- **FICS (Finding Innovative and Creative Solutions) Finalist**, National Science and Technology Park, NUST — the REHABOTICS soft robotic rehabilitation glove placed 8th of 500+ entries, August 2023.

TECHNICAL SKILLS

Methods

Control & Dynamical Systems | System identification; loop shaping; stability margins; PI/PID; lead–lag; notch filtering; MPC (simulation)

Robotics & Mechatronics | Soft pneumatic actuation; electromagnetic/piezoelectric actuator control; sensing–actuation integration; human–machine interfaces

Signal Processing & Machine Learning | Time-domain feature extraction; gradient boosting; XGBoost; CatBoost; SVM; k -NN; autoencoder anomaly detection; real-time inference

Tools

Embedded & Real-Time | STM32 microcontroller; NI myRIO; NVIDIA Jetson Nano; Raspberry Pi 4/5; Arduino; TI TMS320 DSP; embedded C/C++

Instrumentation & Electronics | FRA; oscilloscope; laser displacement sensing; QPD; op-amp signal conditioning; PCB design; LabVIEW; Gwyddion

Design & Simulation | SolidWorks; ANSYS structural FEA; MATLAB/Simulink; Python (NumPy, pandas, scikit-learn, TensorFlow/Keras); 3D printing; silicone casting